

The Simulation Engine

LESSON 02 • TEST & IMPROVE – STUDIO: TEST & SIMULATE

What you'll learn

- Stress test an agent prior to production using the Simulation Engine in Studio.
- Automatically generate test cases and multi-turn conversations based on the agent's goal and role.
- Grade agent responses against ground truth facts using LLM-as-a-judge metrics (task completion and hallucinations).
- Understand automated hardening cycles to auto-repair failing agent configurations.
- Monitor production traffic traces and flag regressions using the Improvement Engine.

Key concepts

Manual chat testing fails to cover the volume and edge cases of real production traffic. Users interact with agents in thousands of unexpected ways. Pre-launch simulation stress tests the agent across diverse scenarios before users encounter them.

The Simulation Engine generates test suites dynamically based on the agent's definition. By analyzing the agent's system prompt, the engine devises multi-turn test conversations targeting typical personas (e.g., policy seekers, detailed researchers, or skeptics).

LLM-as-a-judge metrics evaluate semantic accuracy beyond simple keyword matching. Metrics like task completion and hallucination check whether the output aligns with grounded facts and verify the reasoning in plain English.

Automated hardening automatically optimizes prompts until the agent passes test cases. If a test case fails, the engine runs an optimization loop (up to three times) to automatically adjust the agent's goal and instructions.

The Improvement Engine closes the quality loop by auditing live production traces. After launch, the engine checks real user traces on a schedule, identifies regression issues, and drafts optimized configurations to harden the agent.

In Studio

Studio features: **Safety & Evaluations → Simulation Engine & Improvement Engine.**

1. Go to **Safety & Evaluations → Simulation Engine** in the left menu.
2. Select your agent to clone it into an isolated test environment.
3. Review the proposed test scenarios (e.g., exception handling, user personas) generated by the engine.

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4. Select the scenarios, click **Generate**, and wait for the test suite (e.g., multi-turn conversations) to build.
5. Configure the evaluation by choosing the target test cases, metrics (task completion, hallucinations), and the judge model.
6. Run the evaluation and check the verdict dashboard for task completion scores, hallucination scores, and plain English judge reasoning.
7. Under **Improvement Engine**, register the live agent to audit production traces on a schedule, log critical issues, and apply draft configurations to fix production failures.

TRY THIS

- 1** Go to the Simulation Engine in Studio. Generate a test suite for your agent and review the multi-turn dialogues proposed by the engine.
- 2** Select a subset of test cases, execute the evaluation run, and inspect the judge's reasoning report for any failures.